

YEAR 9 SCIENCE
BIOLOGICAL SCIENCES
REVISION - BODY SYSTEMS

Respiratory System

1. What is the difference between **breathing** and **respiration**?

Breathing - moving air in and out of the lungs.

Respiration - oxidising food to produce energy for cells.

2. (a) Explain how respiration is an example of chemical oxidation.

Food combines with O_2 to produce energy.

- (b) Write a word equation for respiration.



3. Why do cells need oxygen?

To combine with food to give us energy.

4. Why is the blood system so important when considering respiration?

Blood carries food + O_2 to every cell for respiration, and carries wastes out of cells for excretion.

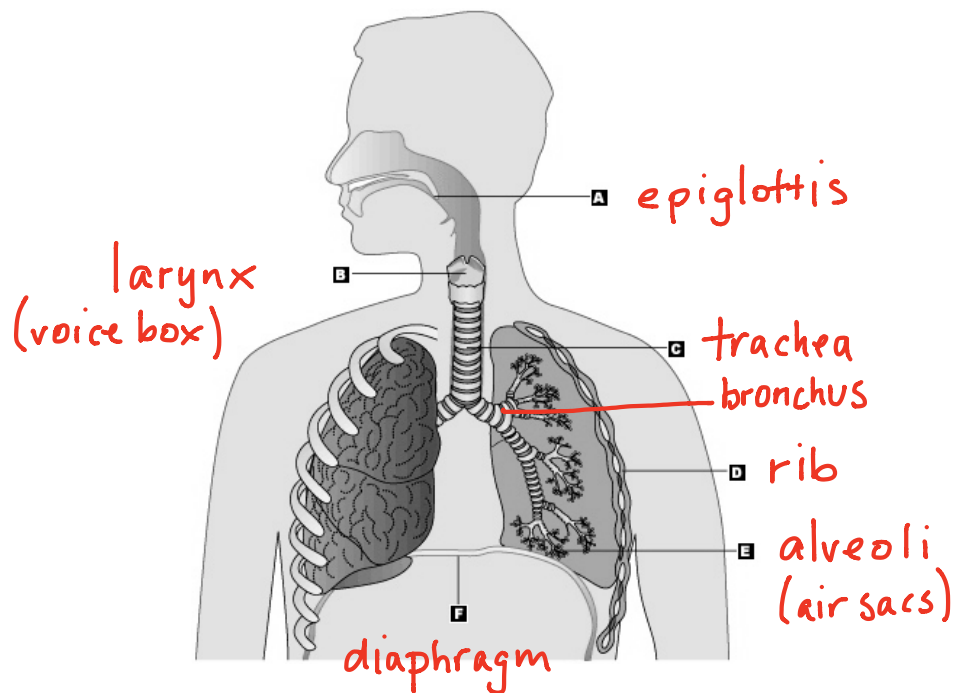
5. Why does an active person need more oxygen than a resting person?

Cells require more energy in an active person so more O_2 is required. Hence the breathing rate increases and the heart beats faster to get O_2 to the cells.

7. Explain the mechanism involved in a human breathing.
i.e. Describe what happens in the body to get air into the lungs and out.

- Rib muscles (intercostal muscles) pull the ribs outwards while the diaphragm moves down.
- Lung volume increases so the air pressure inside is lower.
- Outside air pressure is higher so air rushes into the lungs.
- Rib muscles and diaphragm relax, causing lung volume to decrease.
- Inside pressure is higher, so air rushes out.

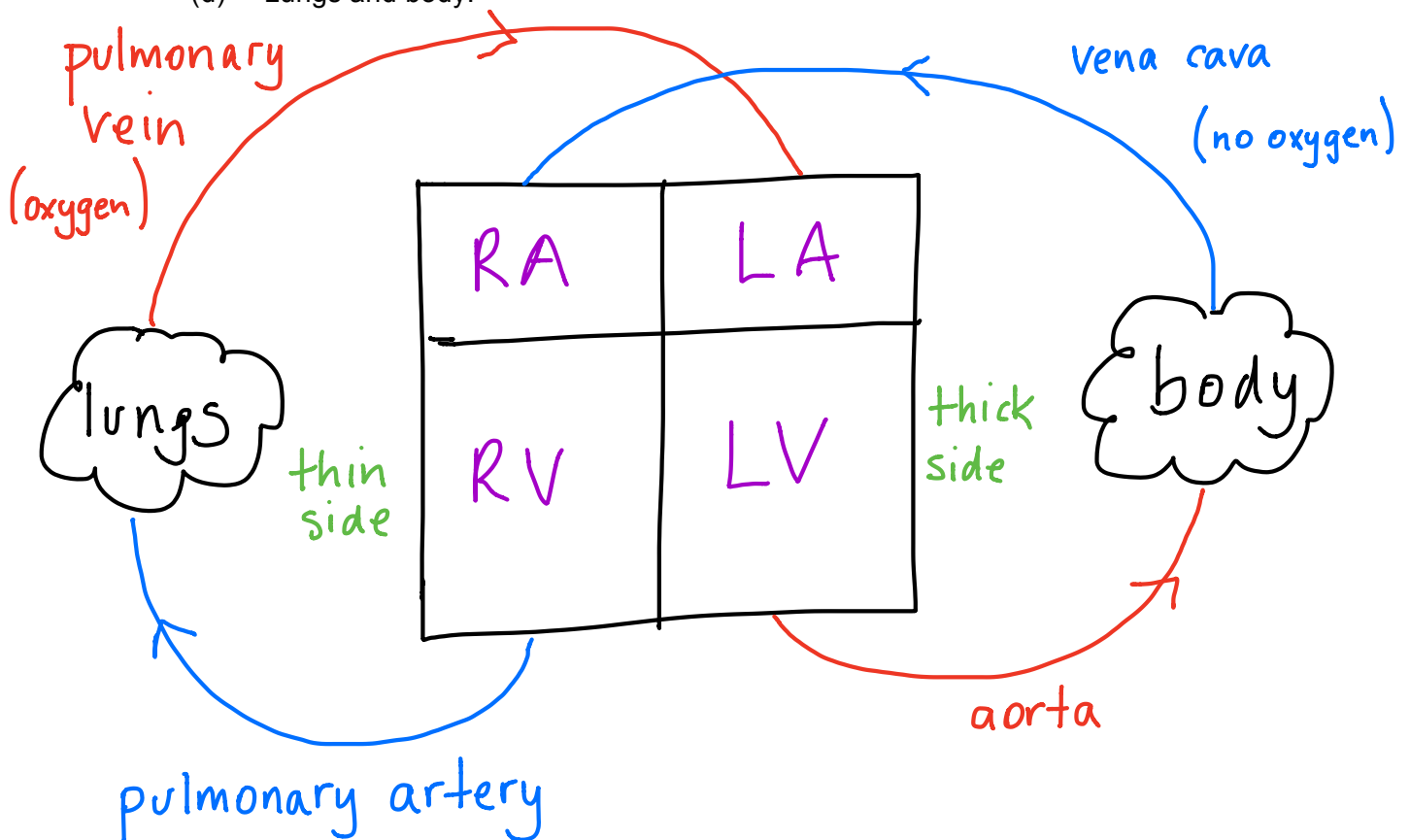
8. Label the following diagram.



Circulatory System

1. Draw a simple labelled diagram to show the blood flow through the heart and around the body. Include the following.

- (a) The arteries and veins, including their names.
- (b) The thick and thin side of the heart.
- (c) Oxygenated and de-oxygenated blood.
- (d) Lungs and body.



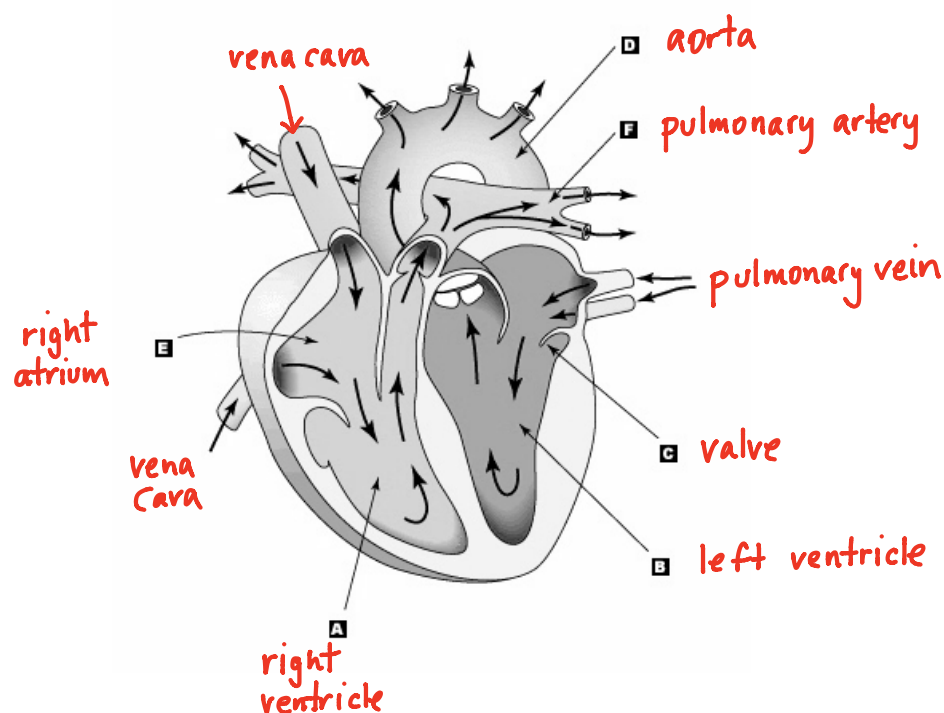
2. Give the function of the following parts of the circulatory system.

- (a) heart pumps blood
- (b) plasma transports O_2 , food, CO_2 , wastes.
- (c) red blood cells carry O_2
- (d) white blood cells attack bacteria
- (e) arteries thick wall, carry blood away from the heart
- (f) veins thin wall, have valves, carry blood back to the heart.
- (g) platelets help clot wounds

3. (a) What is haemoglobin? chemical structure in the blood.

(b) Why is it so important? carries O_2

4. Label the following diagram.

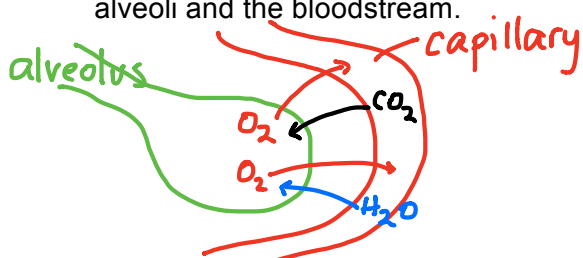


5. (a) What is the role of the valves in the heart?
 Keep blood moving in one direction.
- (b) What is the role of the valves in veins?
 Keep blood flowing in one direction.
- (c) Give **two ways** in which veins are different to arteries.

Veins

- Thin wall.
- Have valves
- Blood pressure lower.

6. Describe how oxygen (and carbon dioxide) diffuses across the membrane between the alveoli and the bloodstream.



- O₂ moves into blood.
- CO₂ + H₂O move into air sac.

7. What advantage is there in having the alveoli so small, and about 300 million of them in a human's lung?

Increases the surface area so far more O₂ and CO₂ can move in and out of the blood.

8. Why does your heart beat faster when you start to run?

Increases blood supply so more food + O₂ reach cells, while wastes get removed faster.

Digestive System

1. (a) What is **digestion**?
 Breakdown of food by chemical + mechanical means.

- (b) Why does food need to be digested?
 Must be very small particles to be absorbed into the blood stream.

- (c) What is **mechanical digestion** and **chemical digestion**?

Mechanical - physical breakdown by the teeth and the muscles of the stomach.

Chemical - Enzymes and saliva breakdown food into small particles that are soluble in water.

2. (a) Explain why organisms need food.

Cells need energy stored in the food and various substances for the maintenance of the body.

- (b) Describe how the digestion of a piece of food containing **starch** takes place from the mouth to the bloodstream.

- Teeth grind the food down.
- Saliva starts to digest the starch.
- Food moves to the stomach.
- Muscles churn the food - some enzymes present.
- Moves to small intestine - enzymes (amylase, protease, lipase) digest food.
- Food particles move through the intestine wall into blood stream.

3. (a) What are **enzymes**?

Chemicals that act as a catalyst to breakdown specific food groups.

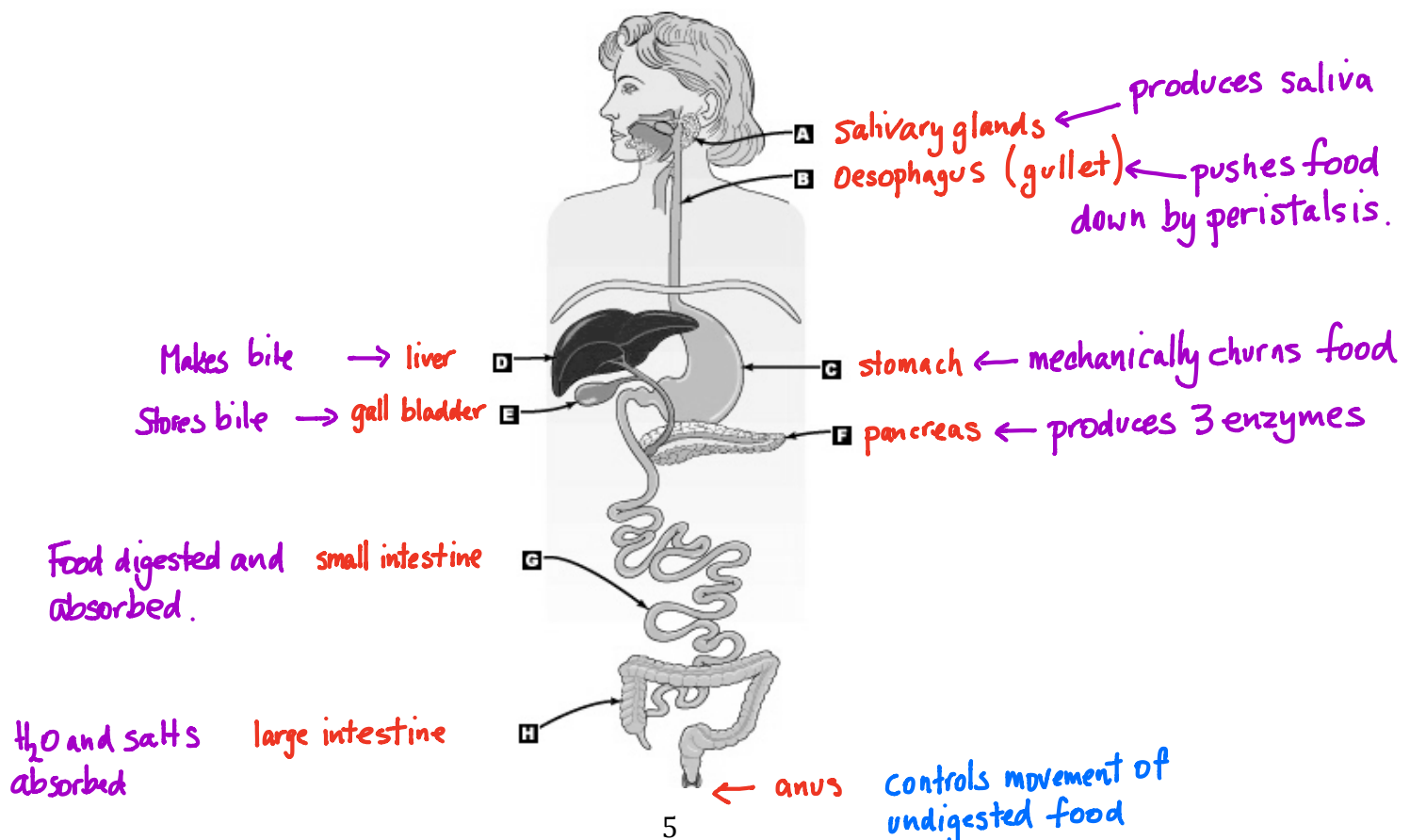
- (b) Name the **three enzymes** in the digestive system and name the foods they target.

Amylase - starch into sugars.

Protease - protein into amino acids.

Lipase - fats and oils.

4. (a) Label the following diagram of the digestive system.



(b) Describe the function of the various parts of the digestive system - consider only the major function. Put them on the diagram above.

5. (a) What are the major nutrient groups?
(b) Describe the major function of these groups.

Carbohydrates - provides energy.

Proteins - repair and growth

Fats + oils - produces 2x energy of carbohydrates.

Vitamins + minerals - used in cell reactions for healthy bones, muscles and nerves.

6. Distinguish between **digestion**, **absorption** and **elimination**.

Digestion - mechanical + chemical breakdown of food.

Absorption - movement of digested particles into the blood.

Elimination - removal of undigested food.

7. (a) You can show signs of malnutrition by eating too much as well as too little food. Explain how this is possible.

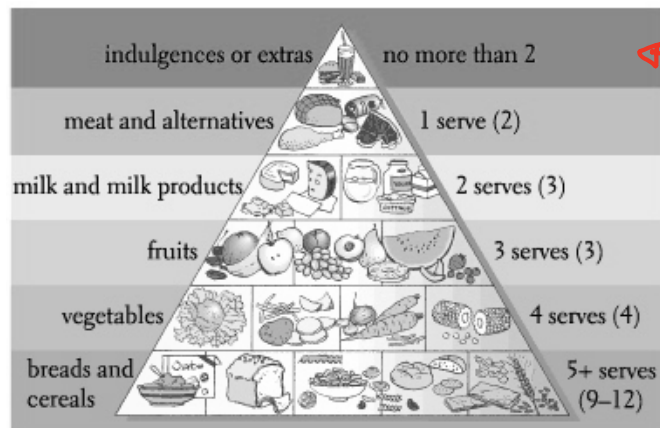
Too much food - obese and overweight.

Too little food - look frail and very thin.

- (b) What is meant by a "balanced diet"?

Have lots of important foods (rice, cereal, fruits, vegetables) and very little of poor foods (fats, oils, sweets).

(c) How does the following diagram relate to having a "balanced diet"?



← Bad foods (little)

← Good foods (lots)

9. What is the role of the **villi**? Increases surface area to absorb digested food quickly.

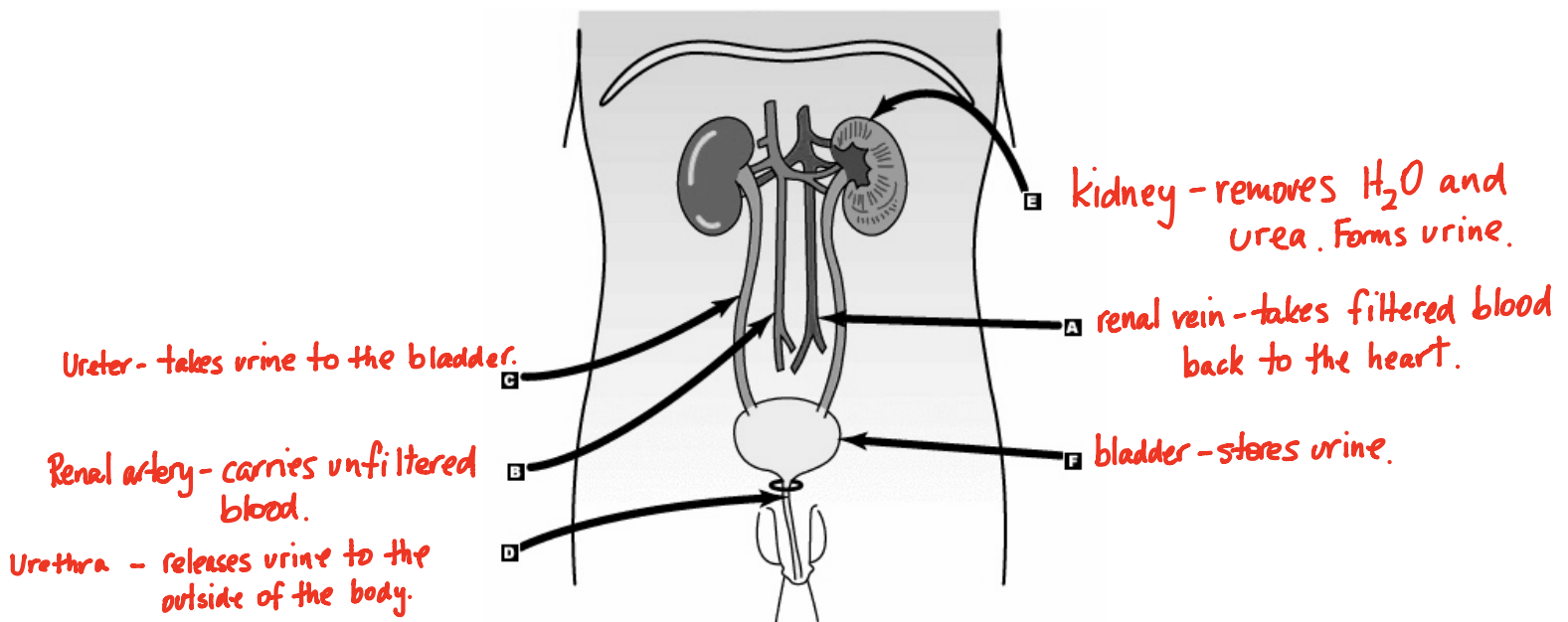
Excretory System

1. What is the difference between **excretion** and **elimination**?

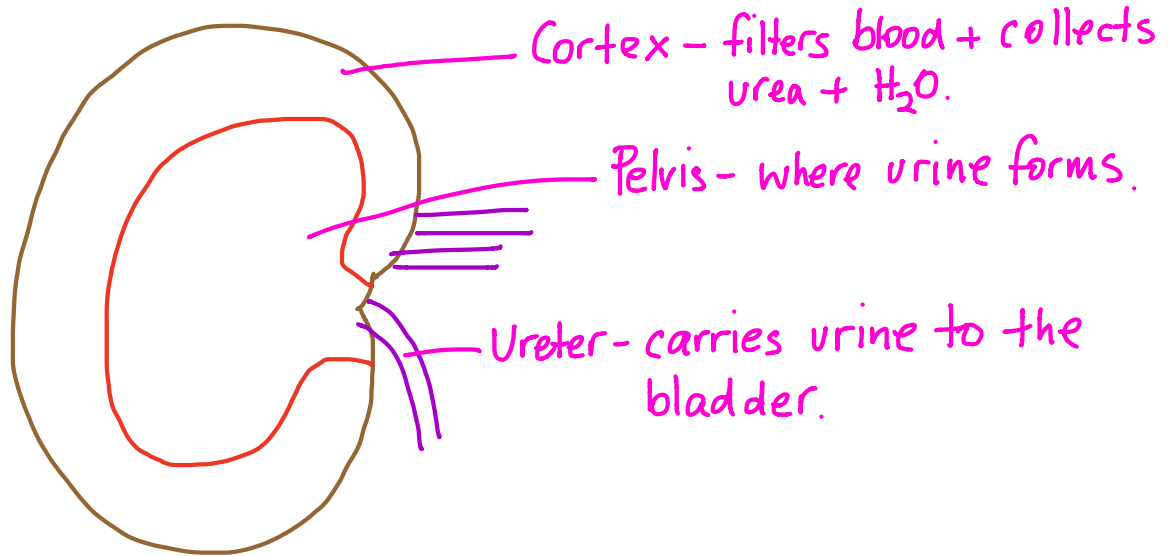
Excretion - removal of wastes (urea, H_2O , CO_2).

Elimination - removal of undigested food.

2. Label the following diagram of the excretory system and indicate the function of each part.



3. Draw a simple diagram of the kidney, label the parts, and describe their function.



4. What are the waste products of excretion?

urea, H_2O , CO_2

5. Name the **organs** responsible for removing wastes from the human body, and indicate which **wastes each removes**.

Lungs - H_2O + CO_2

Skin - H_2O

Kidneys - urea, H_2O

Liver - converts protein wastes into urea.

6. Everyone knows that faeces is a waste! Right or wrong? Explain.

Wrong! It is just undigested food. It does not come from respiration in the cells.